

TRIASHA SARKAR

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[linkedin.com/in/triasha-sarkar](https://www.linkedin.com/in/triasha-sarkar) · github.com/triasha72 · triasha72.github.io/Portfolio

Machine learning engineer and Georgia Tech aerospace MS graduate who builds tested systems from data and evaluation through deployment. Experience spans aircraft-engine diagnostics, time-series ML, retrieval and ranking, recommender systems, GenAI evaluation, scientific ML, and safety-critical engineering.

Education

M.S., Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA Aug 2026

Graduate Research Assistant, Aerospace Systems Design Laboratory (ASDL)

M.Sc., Aerospace Vehicle Design | Cranfield University, United Kingdom Feb 2023

First Class Honours · Thesis on geometry optimization under FAA constraints

B.Tech., Aerospace Engineering | SRM Institute of Science and Technology, India Jun 2021

First Class Honours

Technical Skills

ML & GenAI: PyTorch, scikit-learn, Transformers, RAG, hybrid retrieval, reranking, recommender systems, uncertainty estimation, time-series ML

Systems: Python, SQL, FastAPI, Docker, Kubernetes, GitHub Actions, DuckDB, PostgreSQL/pgvector, Kafka, ONNX, Core ML, Qualcomm QNN

Scientific computing: NumPy, SciPy, pandas, NetworkX, OSMnx, statistical inference, design of experiments, surrogate modeling

Professional Experience

Graduate Research Assistant | Georgia Tech ASDL May 2025 – Aug 2026

- Built a failure-detection workflow for 15,120 simulated rocket-motor time-series cases, reaching 96.8% accuracy and 0.986 ROC AUC on unseen geometries.
- Worked on Delta Air Lines-sponsored HERO retrieval and RAG methods for safety-related information; developed GREEN TEA surrogate models connecting lifecycle, economic, and transportation data.

Machine Learning Engineer | Rolls-Royce Jul 2023 – Apr 2025

- Built predictive-maintenance workflows for aircraft-engine sensor data, including Python pipelines for 1 Hz and 10 Hz multivariate time series, feature engineering, anomaly detection, and model evaluation.
- Worked with Lifecycle Engineers and domain experts on failure modes and repeatable Azure Databricks workflows for engineering datasets.

Intern | Rolls-Royce DataLabs May 2021 – Jul 2021

- Worked on predictive analytics, anomaly detection, exploratory analysis, and proof-of-concept modeling for aircraft-engine sensor data.

Technical Projects

AeroRAG-X | Independent Development of the HERO Retrieval Project 2025 – Present

github.com/triasha72/AeroRAG-X

- Independently extended the HERO retrieval work into a provenance-aware RAG evaluation system with BM25, dense retrieval, hybrid reranking, citation checks, safe agent degradation, and held-out QASPER/SciFact evaluation.
- Measured retrieval behavior from 10K to 1M NASA-text snapshots rather than assuming scale preserved ranking quality.

NewsLens | Recommendation, Ranking, and Production ML Jul 2026

github.com/triasha72/NewsLens

- Built a FastAPI retrieval service with Go ingestion, Kafka, PostgreSQL, Docker/Kubernetes paths, and reproducible MIND evaluation; selected model reached NDCG@10 0.3664.

- Rejected a weaker learned ranker under the frozen comparison and documented application/state-host deployment boundaries.

EdgeGenBench | Scientific ML and On-Device Inference Aug 2026 – Present

github.com/triasha72/EdgeGenBench

- Built an ONNX/QNN deployment-evidence pipeline and evaluated flight anomalies on 17,780 aircraft-disjoint approaches; 94.96% accuracy did not hide the 0.7380 macro-F1 release failure.

AIRFAANS | Coursework Project in Scientific Machine Learning Coursework

github.com/triasha72/AIRFAANS

- Developing and training data-driven surrogate models for airfoil flow fields using the AirfRANS benchmark, with matched comparisons across MLP, graph neural network, and neural-operator approaches.

AeroSynth-Eval | Multimodal AI Evaluation Aug 2026 – Present

github.com/triasha72/AeroSynth-Eval

- Built preference, calibration, transfer, downstream-utility, grouped-uncertainty, and human-alignment evaluation tooling.
- In a ten-seed AGDD study, mixed training improved macro F1 from 0.3881 to 0.4419 but reduced crack recall, so the release gate remains blocked.

IntegrityBench | Integrity and Safety Evaluation Aug 2026 – Present

github.com/triasha72/IntegrityBench

- Built evaluation and release tooling for macro F1, false acceptance, false rejection, calibration, policy citation accuracy, latency, monitoring hooks, and an AWS reference deployment plan.

Surrogate Model Learning | Robust Regression 2025 – Present

github.com/triasha72/Surrogate-model-learning

- Built Gaussian-process and regression workflows with grouped holdouts, operating-domain warnings, uncertainty checks, and data-hashed robustness reports; ten grouped airfoil splits reached mean R^2 0.8662.

Equity Backtest | Walk-Forward Signal Evaluation Jul 2026

github.com/triasha72/Equity-Backtest

- Built a monthly cross-sectional backtest with point-in-time membership validation, no-look-ahead controls, turnover, capacity constraints, and multiple-testing records; return claims remain withheld pending licensed CRSP data.

Project EAGLE | Vehicle Grand Challenge with RIC Enterprise Sep 2025 – Present

Georgia Tech ASDL · Graduate Research Assistant

- Developed a sizing and synthesis tool for the Eco-friendly Airborne Global Luxury Ekranoplan, including passenger layout, cabin geometry in AutoCAD, and vehicle requirements.

Project GREEN TEA | Sustainable Aviation Modeling Sep 2025 – Present

Georgia Tech ASDL · Graduate Research Assistant

- Developed a techno-economic framework for forest-residue sustainable aviation fuel and bioenergy, including jobs, CO₂ impacts, a GREET surrogate, and forest-health modeling.

Additional Experience

Intern | STAR Orbitals Aug 2021 – Oct 2021

Contributed to static-test-pad development for solid rocket motors, Arduino avionics, SolidWorks, Autodesk Fusion 360 mechanical design, and project planning.

Summer Intern | Bangalore Aircraft Industries Pvt. Ltd. Jun 2019 – Jul 2019

Worked on material testing, stress analysis, and structural testing using NASTRAN-PATRAN.

Leadership

Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech 2025 – 2026